

RL-Tuned Synthetic Marine Radar for Sim-to-Real Transfer

Team Name Placeholder

CSCE 775: Deep Reinforcement Learning

I. INTRODUCTION

Marine radar perception and simulation-driven autonomy have received decades of attention, often with a sense of grandeur around fully reliable maritime autonomy and continuous ocean monitoring. However, despite these seemingly steady advances in radar signal processing and synthetic-data pipelines, real radar data remain expensive and sparse, and fixed simulator settings still produce a stubborn sim-to-real gap that limits downstream detection and mapping performance.

This proposal studies whether a reinforcement learning policy can adaptively tune synthetic radar generator parameters to maximize real-data validation performance across environment types (lake, river, coast). The central hypothesis is that environment-conditioned simulator control can outperform fixed or random domain randomization and reduce transfer error under unseen routes and weather/clutter conditions.

The approach is grounded in prior work on domain randomization and adaptive sim-to-real transfer [1]–[13]. However, existing methods either rely on fixed/random randomization, non-RL adaptive tuning, or calibration-centric transfer that is not explicitly conditioned on maritime environment type. Our proposal targets this gap with an environment-aware RL outer loop that directly optimizes real-data downstream task performance, aiming to move beyond seemingly robust simulation results toward genuinely robust field transfer, as summarized in Fig. 1 and contrasted with prior work in Fig. 2.

II. APPROACH

We formulate synthetic radar tuning as a bi-level optimization problem. In the inner loop, a downstream model is trained on synthetic radar generated with parameter vector θ . In the outer loop, an RL policy observes the current generator state, environment-type label, and recent real-validation metrics, then outputs updates to θ .

The action space controls key simulator factors including clutter intensity, noise floor, attenuation, speckle, and target reflectivity variation. The reward is defined as real-data task improvement minus realism/stability penalties to discourage unphysical parameter exploitation. We use Soft Actor-Critic (SAC) as the primary outer-loop optimizer due to sample efficiency and continuous-control support, with TD3 as a backup. A non-RL Bayesian optimization tuner is included as a strong baseline.

Relative to existing adaptive randomization methods (SimOpt, BayesSim, BayRn) [3]–[5], our method makes three

concrete changes: (1) sequential policy-based control of simulator updates rather than one-step or purely inference-driven adaptation, (2) explicit conditioning on maritime environment type ('lake', 'river', 'coast'), and (3) physically bounded action constraints with realism penalties to prevent simulator exploitation. Relative to radar calibration-oriented sim-to-real pipelines [11], the optimization target is not just simulation fidelity but downstream real-task performance under domain holdout.

III. EVALUATION

Evaluation uses real marine radar logs with explicit environment labels and strict split protocols by route/location to prevent leakage. We compare: (1) fixed synthetic defaults, (2) uniform random domain randomization, (3) Bayesian optimization tuning, (4) RL tuning without environment conditioning, and (5) RL tuning with conditioning.

Primary metrics are downstream real-data task score and relative sim-to-real gap reduction. Secondary metrics include cross-domain generalization (train on two environment types, test on held-out type), unseen-location robustness, seed variance, and performance under low-SNR/high-clutter stress conditions. We will report both best-case performance and sample-efficiency measured by real-validation calls required to reach target accuracy.

To explicitly support the novelty claims, we add claim-linked evaluations: (a) sequential adaptation benefit by comparing RL vs BayRn under matched real-evaluation budgets [5]; (b) environment semantics benefit by comparing conditioned vs unconditioned RL policies; and (c) plausibility benefit by reporting parameter-bound violations and instability with/without realism regularization. These analyses directly test how our method improves over existing randomization and simulator-adaptation approaches [1]–[4], [6], [7].

Published results already show clear failure modes for non-adaptive baselines (Fig. 3). In real FMCW radar transfer, simulation-only baselines can collapse to near-chance balanced accuracy (about 50% for occupancy and 33% for people-counting), while calibrated adaptation substantially improves performance (about 97% and 72%, respectively) [11]. In synthetic-to-measured SAR transfer, synthetic-only and simple-noise baselines are consistently weaker than stronger adaptation strategies across multiple architectures [10]. These patterns motivate the need for a more structured, environment-aware adaptation controller rather than fixed randomization.

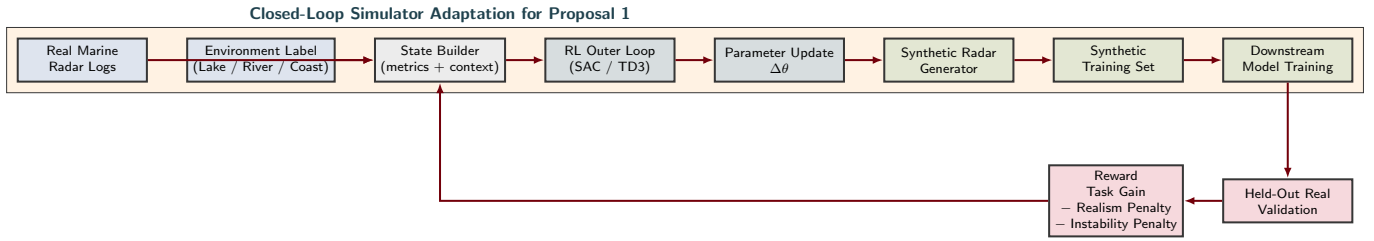


Fig. 1. Proposal 1 closed-loop method: an RL outer loop updates simulator parameters from real-validation reward, conditioned on environment type.

Method Family	Real-Feedback Adaptive	Env-Type Aware	Sequential Policy Control	Physical Bounds	Task-Reward Optimized
Fixed Domain Randomization (Tobin, Peng)	✗	✗	✗	✗	✗
SimOpt / BayesSim	✓	✗	✗	✗	✗
BayRn / ADR / CDR	✓	✗	✗	✗	✗
Radar Calibration (Digital Twin style)	✓	✗	✗	✓	✗
Proposed Method (Proposal 1)	✓	✓	✓	✓	✓

Fig. 2. Related-work capability comparison and the explicit improvement axes targeted by Proposal 1.

IV. CONCLUSION

This project targets a practical and research-relevant problem: improving sim-to-real transfer for marine radar perception through adaptive synthetic data generation. The expected contribution is an RL-controlled simulator tuning framework that is environment-aware, benchmarked against strong non-RL alternatives, and evaluated under realistic transfer stress tests. If full RL tuning is unstable, a contextual-bandit or Bayesian fallback will still provide a complete, meaningful project outcome.

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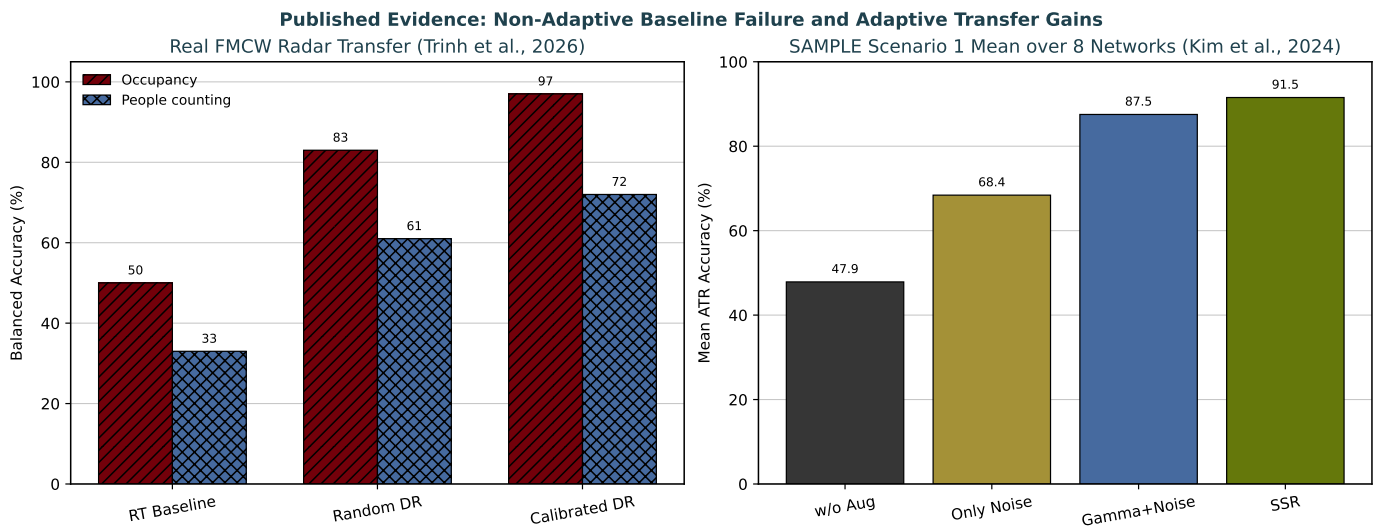


Fig. 3. Published quantitative evidence motivating Proposal 1. Left: non-adaptive and weakly adaptive baselines can fail under sim-to-real radar transfer; calibrated/adaptive methods improve substantially. Right: synthetic-only and simple noise baselines underperform on measured-data transfer compared to stronger adaptation strategies.